

Appendix A. Manifold Geometry of SPD Matrices

Before introducing our regression model, we briefly review the geometric structure of the manifold of symmetric positive definite (SPD) matrices under the affine-invariant Riemannian metric[1, 2, 3].

We consider the space $Sym^+(d)$, which denotes the Riemannian manifold of all $d \times d$ symmetric positive definite matrices, and the dimension of space $Sym^+(d)$ is $d(d+1)/2$. We use the affine-invariant Riemannian metric (AIRM) , the tangent space at a point $p \in Sym^+(d)$ is given by

$$T_p Sym^+(d) = \{S \in \mathbb{R}^{d \times d} \mid S = S^T\}, \quad (A.1)$$

and the Riemannian metric is defined as

$$\langle U, V \rangle_p = \text{tr}(p^{-1}U p^{-1}V), \quad U, V \in T_p Sym^+(d). \quad (A.2)$$

This metric induces a geodesic distance between any two SPD matrices $p, q \in Sym^+(d)$, given by

$$d_{Sym^+(d)}(p, q) = \|\log(p^{-1/2}qp^{-1/2})\|_F, \quad (A.3)$$

where $\log(A) = -\sum_{k=1}^{\infty} \frac{(I-A)^k}{k}$ and $\|\cdot\|_F$ is the Frobenius norm. To move along geodesics on $Sym^+(d)$ from a given base point p , we introduce the exponential map, which plays a role analogous to vector addition in Euclidean space. Specifically, the exponential map at p maps a tangent vector $V \in T_p Sym^+(d)$ to the point on the manifold reached by following the geodesic starting from p in the direction V ,

$$\text{Exp}(p, V) = p^{1/2} \exp(p^{-1/2}V p^{-1/2}) p^{1/2}, \quad (A.4)$$

where $\exp(A) = \sum_{k=0}^{\infty} \frac{1}{k!} A^k$. Correspondingly, the logarithm map provides the inverse operation. Given two points $p, q \in Sym^+(d)$, the logarithm map at p returns the tangent vector $\text{Log}(p, q)$ that points from p toward q along the geodesic. In this sense, $\text{Log}(p, q)$ plays a role analogous to vector subtraction in Euclidean space. More specifically, the logarithm map on the SPD manifold admits the closed-form expression,

$$\text{Log}(p, q) = p^{1/2} \log(p^{-1/2}qp^{-1/2}) p^{1/2}, \quad (A.5)$$

where $\log(\cdot)$ denotes the matrix logarithm.

Since tangent vectors attached to different points on the SPD manifold belong to different tangent spaces, they cannot be directly compared. Therefore, parallel transport is required to map them into a common tangent space while preserving their geometric meaning. Let $\mathcal{M}$ be a differentiable manifold endowed with an affine connection ∇ , and let I be an open interval. Let $c : I \rightarrow \mathcal{M}$ be a differentiable curve on $\mathcal{M}$, and let V_0 be a tangent vector in $T_{c(t_0)}\mathcal{M}$, where $t_0 \in I$. Then, there exists a unique vector field $V(t)$ along c such that $V(t_0) = V_0$ and $\nabla_{\dot{c}(t)}V(t) = 0$. The vector field $V(t)$ is called the parallel vector field along c , and $V(t_1)$ is called the parallel transport of V_0 from $c(t_0)$ to $c(t_1)$ along c . We denote this transport by $\Gamma_{c(t_0) \rightarrow c(t_1)}(V_0)$.

Appendix B. Intrinsic Derivation of the Proposed Dimensionality Reduction under the Affine-Invariant Riemannian Metric

Appendix B.1. Affine-Invariant Riemannian Metric on the SPD Manifold

Let $Sym^+(d)$ denote the manifold of $d \times d$ symmetric positive definite (SPD) matrices. This manifold admits a canonical affine-invariant Riemannian metric (AIRM), which has been widely used in statistical analysis of covariance matrices.

Under this metric, the geodesic distance between $p, q \in Sym^+(d)$ is defined as:

$$d_{\text{AIRM}}^2(p, q) = \left\| \log \left(p^{-\frac{1}{2}} q p^{-\frac{1}{2}} \right) \right\|_F^2. \quad (\text{B.1})$$

This distance is determined by the congruence transformation $p^{-\frac{1}{2}} q p^{-\frac{1}{2}}$, which characterizes the intrinsic geometric relationship between p and q under the affine-invariant structure. This transformation can be viewed as a relative representation of q with respect to the reference point p . Particularly, the interaction between two SPD matrices can be described through the congruence-invariant quantity $p_{ij} = p_i^{-\frac{1}{2}} p_j p_i^{-\frac{1}{2}}$. Such a representation has been employed in the analysis of dynamical brain functional connectivity[2].

Appendix B.2. Objective Function Derivation

Let $\{p_i\}_{i=1}^N \subset Sym^+(d_1)$ be a collection of SPD matrices, and let $\mathcal{S}_{d_1, d_2} = \{B \in \mathbb{R}^{d_1 \times d_2} : B^T B = I_{d_2}\}$ denote the Stiefel manifold. Based on the affine-invariant Riemannian structure introduced above, we consider the intrinsic geometry of $Sym^+(d_1)$ relative to the identity matrix as a common base

point. In this case, the representation reduces to $d(I, p_i) = I^{-1/2} p_i I^{-1/2} = p_i$. Hence, the intrinsic structure relative to the identity is fully encoded by the matrices $\{p_i\}$ themselves.

To perform dimensionality reduction, we introduce an orthogonal projection matrix $B \in \mathcal{S}_{d_1, d_2}$ with $B^T B = I_{d_2}$ and define the low-dimensional representation $q_i = B^T p_i B \in \text{Sym}^+(d_2)$. In order to compare the projected matrices with the original ones within the same ambient space, we reconstruct them as $\hat{p}_i = B q_i B^T = B B^T p_i B B^T$. Following [2], the structural distortion is measured by the Frobenius discrepancy between the original and reconstructed matrices:

$$\begin{aligned}
E(B) &= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_i (\mathrm{d}_{\text{Sym}^+(d_1)}(I_{d_1}, p_i) - \mathrm{d}_{\text{Sym}^+(d_1)}(I_{d_1}, \hat{p}_i))^2 \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_i \left\| I^{-\frac{1}{2}} p_i I^{-\frac{1}{2}} - I^{-\frac{1}{2}} \hat{p}_i I^{-\frac{1}{2}} \right\|^2 \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_i \|p_i - B q_i B^T\|^2 \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_i \|p_i - B B^T p_i B B^T\|^2 \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_{i=1} \operatorname{tr} \left((p_i - B B^T p_i B B^T)^2 \right) \\
&= \operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} \sum_{i=1} \left[\operatorname{tr}(p_i^2) - \operatorname{tr}(B^T p_i B B^T p_i B) \right].
\end{aligned}$$

Note that the term $\sum_{i=1}^N \operatorname{tr}(p_i^2)$ does not depend on B and is therefore constant with respect to the optimization variable. Consequently, it does not affect the location of the minimizer. The optimization problem is thus equivalent to minimizing $-\sum_{i=1}^N \operatorname{tr}(B^T p_i B B^T p_i B)$ over $B \in \mathcal{S}_{d_1, d_2}$, which is in turn equivalent to maximizing $\sum_{i=1}^N \operatorname{tr}(B^T p_i B B^T p_i B)$ on the same manifold. Hence, the dimensionality reduction problem reduces to

$$\operatorname{argmin}_{B \in \mathcal{S}_{d_1, d_2}} - \sum_{i=1}^N \operatorname{tr}(B^T p_i B B^T p_i B) = \operatorname{argmax}_{B \in \mathcal{S}_{d_1, d_2}} \sum_{i=1}^N \operatorname{tr}(B^T p_i B B^T p_i B).$$

This formulation explicitly preserves the distances from each SPD matrix to the identity matrix, ensuring that the low-dimensional projections

maintain the essential geometric structure of the original data.

Appendix B.3. Riemannian Gradient on the Stiefel Manifold

To optimize the proposed dimensionality reduction objective on the Stiefel manifold, we first derive the Euclidean gradient of $f(B) = \sum_{i=1}^N \text{tr}((B^T p_i B)^2)$ for $B \in \mathcal{S}_{d_1, d_2}$. Letting $c_i = B^T p_i B$, the differential of c_i with respect to B is $dc_i = (dB)^T p_i B + B^T p_i (dB)$, which yields the differential of $f(B)$:

$$\begin{aligned} df(B) &= 2 \sum_{i=1}^N \text{tr}(c_i dc_i) \\ &= 2 \sum_{i=1}^N \text{tr}(c_i (dB)^T p_i B) + 2 \sum_{i=1}^N \text{tr}(c_i B^T p_i (dB)) \\ &= 2 \sum_{i=1}^N \text{tr}((p_i B c_i + p_i B c_i)^T dB) \\ &= 4 \sum_{i=1}^N \text{tr}((p_i B B^T p_i B)^T dB). \end{aligned}$$

Hence, the Euclidean gradient is $\nabla f(B) = 4 \sum_{i=1}^N p_i B B^T p_i B$. To obtain the Riemannian gradient of $f(B)$ on the Stiefel manifold $\mathcal{S}_{d_1, d_2}$, we project the Euclidean gradient $\nabla f(B)$ onto the tangent space at B :

$$\text{grad } f(B) = \nabla f(B) - B \text{sym}(B^T \nabla f(B)),$$

where $\text{sym}(X) = \frac{1}{2}(X + X^T)$ denotes the symmetric part of a matrix. This projected gradient is then used in the Manopt toolbox to perform optimization on the manifold.

References

- [1] Kim, H.J., Adluru, N., Collins, M.D., Chung, M.K., Bendlin, B.B., Johnson, S.C., Davidson, R.J., Singh, V., 2014. Multivariate general linear models (MGLM) on Riemannian manifolds with applications to statistical analysis of diffusion weighted images, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2705–2712.

- [2] Wang, M., Wang, Y., Yang, Y., 2025. Dynamic and low-dimensional modeling of brain functional connectivity on riemannian manifolds. *NeuroImage* , 121243.
- [3] Xiong, D., Ying, S., Zhu, H., 2022. Intrinsic partial linear models for manifold-valued data. *Information Processing & Management* 59, 102954.